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The Guide to Optimizing Data Protection for the Hybrid Cloud

Why and How to Optimize it Now for IBM Power Environments

Most organizations today are running hybrid cloud environments. The pace of business and the macro-economic climate add to the mix of new challenges for organizations. With limited resources and expanding landscapes, the challenges are multiplying and protecting an organization's data is evolving as well. So how do you optimize your data protection for today's hybrid cloud environments?

WHY OPTIMIZE?

The first question may be why optimize? Your data backup and recovery infrastructures may have been running stably for years. However, common dynamics and some nuances from supporting hybrid cloud environments introduce some new challenges that should not be overlooked.

Security

Cybersecurity and ransomware attacks are the latest challenges for all organizations. Keeping up with the latest attacks with limited staff can be time-consuming and take away from critical projects and day-to-day operations. Have you safeguarded your on-premises data enough? What about your data that resides in the cloud? How is that protected? How can you manage both?

Growing data

Data growth is a common reality. Today's data protection scenarios include protecting data on-premises and in the cloud, but optimization is always key when planning for growth. How can you manage and minimize the storage footprint of your protected data both on-premises and the cloud?

Performance

Faster backups and more importantly, faster recovery has always been the cornerstone of data protection. The fast-paced ecommerce world of high-level transactions only exasperates the problem. IBM Power is purposely built for heavy transaction loads. Optimizing the performance of data protection to match the needs becomes today's imperative.

Efficiency

Organizations should be asking what more they can do to protect the data not only on-premises but in the cloud too. Is there more to protect in the cloud? How easy is it to do that? Cloud vendors often charge not only for the storage footprint in the cloud, but the traversing network traffic on the cloud to move the data as well. How do you keep the transmission of data to a minimum to keep your networks free for important transactions

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and to keep cloud network charges at a minimum? It's just another part of optimizing data protection for hybrid environments.

Simplicity
With the growing complexity of environments, the reality of limited resources to keep up to date with new applications and innovative technology continually resonates with most organizations. With IT staff already stretched, keeping systems up to date and running efficiently for internal and external customers is often overlooked.

SO HOW DO YOU OPTIMIZE?

The biggest challenge is trying to optimize without disrupting your existing backup and recovery infrastructure and without introducing new processes or techniques that require new training for the limited resources on hand. Keeping disruptions at bay should be a part of the optimization process.

Many organizations are discovering the secret of leveraging the power of VTL (virtual tape libraries). VTL technology is not just about tapes anymore. VTL is a storage virtualization technology. By virtualizing any storage to disk or cloud, the solution is perfect for optimizing data protection for hybrid cloud environments. In the past, VTL was popular to help organizations migrate from using tape for backup and recovery to disk, by mimicking existing physical tape libraries to more cost-effective disks by virtualizing the storage and not disrupting current infrastructures. Data that was previously backed up to physical tape libraries were backed up to disk storage instead using VTL technology. It can do the same too for migrating to the cloud and optimizing protection of hybrid cloud environments.

THE POWER OF VTLS

VTLs are excellent data movers. What does that mean? They can move data between locations seamlessly, whether that is from tape to disk, disk to disk, disk to cloud, tape to cloud, or cloud to cloud. That is one of the fundamental powers of VTLs. Let's consider the many benefits of VTLs today to optimize data protection for hybrid cloud environments and for meeting the challenges of today.

No changes to existing backup and recovery processes
One of the biggest benefits continues to be that no changes are required to existing backup and recovery software, processes, and infrastructures in using VTLs. Seamless integration is still one of the benefits that really

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stand out in eliminating any disruption of service. VTLs work with existing backup and recovery environments. No changes to existing processes save on costs and enables organizations to leverage existing skill sets. FalconStor StorSafe VTL can be implemented without changes to existing backup and recovery processes.

Protects against ransomware

FalconStor StorSafe VTL protects against ransomware with immutable backups. An immutable backup secures data by making it non-rewritable and non-erasable. As a result, it provides extra security by prohibiting accidental data deletion or ransomware attacks. Once data is written, it cannot be altered or overwritten by some virus, ransomware, or other malicious software.

Immutability ensures that your data remains safe and recoverable at any time. To comply with data security standards, FalconStor StorSafe VTL offers backup immutability features. Air-gapped and immutable backups protect against cyber and ransomware attacks with provisioning Write-Once-Read-Many (WORM) virtual tapes and integration with IBM Immutable Cloud Object Storage to protect IBM Power cloud environments. StorSafe VTL encrypts data in motion and at rest using AES-256 with StorSafe VTL managed keys. These features adhere to widely recognized governmental and industry requirements for compliance.

Accelerates backups and recovery

Ever-increasing volumes of data end up bringing a constant worry that backups will not be completed within the scheduled timeframe. Leveraging a VTL for backups and recovery can increase speeds from 2x to 10x with FalconStor StorSafe VTL. Flexible inline or post processing deduplication options are built in to help with optimizing performance to meet backup window demands. Disk based access and built in deduplication algorithms minimize data transmission for faster recovery.

Minimizes storage footprint

Patented deduplication technology in StorSafe VTL can reduce the size of backups by up to 95%, significantly reducing storage costs and bandwidth costs in the cloud. StorSafe VTL supports global deduplication with a single instance repository making it the perfect solution for hybrid cloud environments.

Deduplication has been a method for saving on storage costs for many years. By eliminating duplicate copies during the data protection process, organizations can save on storage costs by minimizing the space and capacity required to store data. The cost savings are often very substantial, especially when you multiply the deduplication

methodologies across replicating backups for disaster recovery (DR) and reducing the storage footprint.

Lowers storage costs
By utilizing less storage with a smaller storage footprint leveraging deduplication, it inherently saves on overall storage costs. Tape archives can be consolidated as well to the cloud in a hybrid cloud environment easily leveraging StorSafe VTL and offering even more savings in storage and management costs. Many organizations, like Tupperware, have migrated archived data on tape to the cloud, such as IBM Power Virtual Server (PowerVS), leveraging StorSafe VTL for the migration.

Another benefit is the substantial savings that stem from network bandwidth. With storage data being deduplicated, the amount of data for transmission is significantly reduced and therefore reduces the traffic across the network as part of the data protection process. With less data replicated after deduplication, less band width is used. Less network transmissions are an added cost savings, especially for the cloud where vendors also charge for network utilization. With less data having to traverse the network, network utilization costs are reduced in the cloud environment.

Simplifies cloud migrations
Backup on-premises, then restore to the cloud –it is that simple! VTLs provide more than optimizing your backups to disk vs. only to tape. As primary data movers, the power of VTLs helps in moving the data efficiently, taking advantage of the inherent benefits of the deduplication and more built right in. As hybrid cloud organization progresses, cloud migrations will continue to be inevitable.

Optimizes disaster recovery
Virtual tape libraries have been used to optimize backup and recovery by providing dependable disaster recovery and hybrid cloud backup and recovery. Virtual tape libraries can move data easily and replicate efficiently, making them a perfect tool for protecting data in the cloud and on-premises. StorSafe VTL is certified to protect IBM PowerVS cloud implementations.

Facilitates tape management
The advantage of VTLs is the ability to also manage tapes. Hybrid cloud environments usually involve a mix of protected data on disk and on tape. The tape management capabilities of StorSafe VTL are just another example of the power of VTLs with their traditional use and beyond.

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WHY FALCONSTOR STORSAFE VTL?

FalconStor StorSafe VTL dominates the VTL market and has many implementations that demonstrate the power of VTLs. Today, VTLs can be leveraged for much more in hybrid cloud environments, and importantly, StorSafe VTL is certified by IBM for IBM Power environments with IBM i, AIX, or Linux. StorSafe VTL is certified by IBM to protect IBM Power Virtual Server (PowerVS) cloud environments, a key component for IBM hybrid cloud environments.

Leverage the power of VTLs for your hybrid cloud environment today.

Contact FalconStor: https://www.falconstor.com/contact

An IBM-certified version for PowerVS, StorSafe VTL for PowerVS, is fully integrated into the IBM Cloud Catalog with one-click provisioning and integrated billing with PowerVS.

IBM Partnership: https://www.falconstor.com/partners/ibm_partnership/

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